xptNo: 14

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Write a Code using RAW Sockets to implement packet Sniffing.

Aim:

To implement a Simple packet Sniffer in Java that captures and displays network packets.

Algorithm:

- Start the program
- Create a RAW Socket using Socket (AF_PACKET, SOCK_RAW.)
- Bind the Socket to the required network interface.
- Continuously receive packet from the network.
- Extract the Ethernet header from each packet
- Identify the protocol type, Such as IPV4 ARP or IPV6
- IPV4 packets, extract the Source and destination IP address.
- Display the packet information.
- Repeat until the user terminates the program.
- close the Socket.

Program:

```
    import Socket
    import Struct

    raw_Socket = Socket.Socket ( Socket.AF_PACKECT , Socket.
                                 Sock_RAW, Socket.ntohs(3))

    Print (" Packet Sniffer started ...")

    Print (" Waiting for packets ...\n")
```

Sample Outcome:

```
packet Sniffer Started
waiting for packets

Source MAC : 3c : 52 : 82 :11 : 22 : 33
Destination MAC : ff : ff : ff : ff : ff : ff : ff
protocol : 0x 806
Packet Type : ARP
- - - - - - - - - - - - - - - - - -
Source MAC : 3c : 52 : 82 : 11 : 22 : 33
Destination MAC : 84:16 :19 :44 : 55 : 66
protocol : 0x800
Source IP : 192.168.1.10
Destination IP : 192.168.1.1
TTL : 64
IP protocol : 6
- - - - - - - - - - - - - - - - - -
```

```
try:
    While True :
        packet , address = raw_Socket . recfrom (65535)
        ethernet_header = packet [1 :4]
        dest_mac , Src = mac, protocol = Struct . unpack ("! 6s6sH",
                                        ether header )

        dest_mac = ":". join ( "%. 02x" %. byte for byte in dest_ma

        Src_mac = ":".join ( "%.2x" %. byte for byte in Src_mac )

        print (" = = = = = = = = = = = = = = = = = = = = ")

        print (" Source MAC :", Src_mac)

        print ("Destination MAC :", dest_mac)

        print (" protocol :" , hex (protocol))

        if protocol == 0x0800 :
            ip_header = packet [14 ; 34]

        version _ihl , tos, total_length, identification, \ flags _ fragmen
                ttl , proto, checksum , \ Src_ip , dest_ip = Struc
                        . unpack ("! BBHHHBBH4s4s", ip_header)

        ~~Src~~ ip , ~~dest ip~~ = ~~Struct~~ . ~~unpack~~ & ~~Src~~'

        Src_ip = Socket. inet _ntoa (Src _ip)
        dest_ip = Socket .inet_ntoa (dest_ip)
    elif protocol = 0x0806:
            print ("packet Type : ARP")

    elif protocol== 0x 86DD:
            print ("packet Type : IP V6")

    ~~elif~~ else:
            print ("packet Type : other")

  except Keyboard Interrupt:
              print ("\n packet Sniffing Stopped.")

  finally:
         raw_Socket.close()
```

Result:

Thus, a packet-Sniffing program using RAW Socket was Successful implemented. The program Captures network packets & display important information Such as Source MAC address, destination MAC address, Source IP address, destination IP address and TTL